

Introduction & Motivation

Portable, low-cost Brain–Computer Interface (BCI) devices now make it possible to observe neural activity in near real time, well outside the laboratory.

In classrooms, student engagement is still mostly assessed through observation or questionnaires, which are subjective and retrospective. EEG offers a complementary continuous, physiological signal that can be related to specific moments of a lesson.

This work presents a classroom prototype that collects and visualizes engagement from several students at once, focusing on technical feasibility and practical deployment rather than clinical precision.

Contributions

- A distributed, low-cost acquisition model: one EEG headset and one Raspberry Pi per student.
- A per-student engagement estimate streamed in real time over the local network.
- A single teacher interface that unifies live monitoring, lesson presentation, and post-lesson review.
- A design that treats engagement as a supportive indicator, not an automatic score.

Background

- EEG activity in the theta, alpha, and beta bands tracks attention and mental effort during learning [1, 5].
- Even single-channel setups give useful attention features at lower cost [3].
- Engagement is commonly estimated with the beta-to-alpha-theta ratio [4].
- Portable EEG is sensitive to movement and electrode contact, so values should be read as indicators, not measurements.

System at a Glance

Layer	Implementation
Acquisition	Unicorn EEG + Raspberry Pi
Transport	Lab Streaming Layer (LSL)
Processing	FFT band power, $\beta/(\alpha+\theta)$
Interface	Streamlit teacher dashboard

Teacher Interface & Workflow

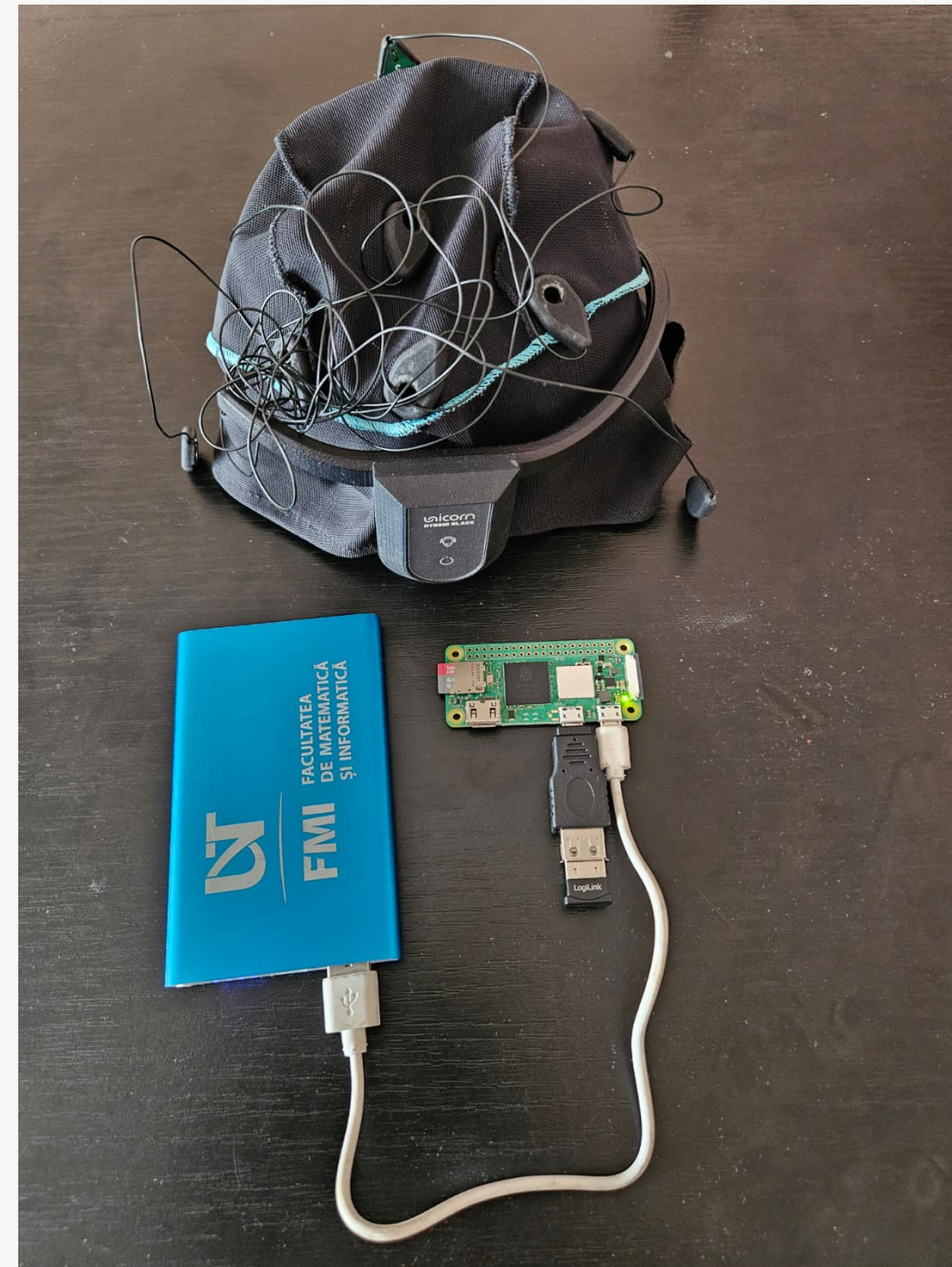
A Streamlit application organized around the teaching workflow, with four areas: lesson creation, teaching, reporting, and administration.

- Lesson creation: upload material, deduplicate by file hash, generate PDF previews, extract speaker notes.
- Teaching: in-app slides with presenter view, keyboard navigation, and a separate audience window.
- Live monitoring: compact student cards (engagement value, stream name, freshness) plus a detailed inspection view;
- Reporting: engagement by lesson segment, as a chart and a table for post-lesson review.

System Architecture

The prototype is a distributed classroom system: N student stations and one teacher workstation. Each station pairs a Unicorn headset with a Raspberry Pi node that publishes the EEG signal as an LSL stream. The workstation discovers the streams, computes engagement, and renders the dashboard.

Responsibilities are split into four parts — acquisition, transport, processing, and storage — so adding a student means adding a node, without changing the teacher application.



A student station: Raspberry Pi, power bank, and Unicorn EEG headset.

Why LSL?

Lab Streaming Layer provides timestamped streaming and automatic discovery on the local network, so nodes and the teacher application can start in any order [2].

Signal Processing Pipeline

The processing chain is deliberately lightweight and runs on the teacher workstation; the Raspberry Pi only acquires and publishes samples.

1. A background collector reads each LSL stream into a per-stream state.
2. Samples are validated and buffered into short windows.
3. The window mean is removed; an FFT estimates power in the θ , α , and β bands.
4. The engagement value follows the Pope ratio [4]:

$$E = \frac{P_{\beta}}{P_{\alpha} + P_{\theta} + \varepsilon}$$

where $\varepsilon = 10^{-6}$ avoids division by zero. The ratio is clipped, nonlinearly scaled, smoothed, and calibrated before display.

The short path keeps the link between raw samples and the displayed value easy to explain, and the raw signal history stays available for inspection.

Results

The end-to-end system was evaluated in a single 20-minute session with five participants, each on their own headset and Raspberry Pi node, all streaming at once. The application recorded engagement per slide and produced the report automatically.

- Overall engagement was around 70%: high in the first part of the lesson and clearly lower in the denser second half.
- The per-student view captured individual differences — most stayed engaged throughout, while one participant disengaged in the second half.
- Nodes operate independently; one failing node does not stop the system.

Observed limitations

Sensitivity to movement artifacts and electrode positioning, and variability between users. A single small session shows feasibility, not general results.

Conclusion & Future Work

A low-cost classroom neurofeedback system can be built from accessible hardware and open software, combining acquisition, streaming, processing, and a teaching interface in one tool.

Next steps: larger classroom studies, quantitative evaluation, improved preprocessing, and the integration of additional physiological signals.

Source Code & Full Thesis



<https://github.com/Relu12345/Dizertatie>

References

- [1] A. Apicella et al. Eeg-based measurement system for monitoring student engagement in learning 4.0. *Scientific Reports*, 12(1), 2022.
- [2] C. Kothe et al. The Lab Streaming Layer for synchronized multimodal recording. *Imaging Neuroscience*, 3, 2025.
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- [4] A. T. Pope, E. H. Bogart, and D. S. Bartolome. Biocybernetic system evaluates indices of operator engagement in automated task. *Biological Psychology*, 40(1–2):187–195, 1995.
- [5] A. T. Poulsen et al. EEG in the classroom: Synchronised neural recordings during video presentation. *Scientific Reports*, 7, 2017.